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Project Title:	Heavy Ion Microbeam and Micron Resolution Detector		
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Zip Code:	11973	**Congressional District:**	1
Comments:			
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Contact Monitor:		**Contact Phone:**	
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Task Description:	(1) Aims of the Project: Background and Significance. The use of microbeam provides a unique way to control the number of particles traversing individual cells and localizing the dose within the cell. High energy heavy charged particles transfer their energy to biological organisms through high density ionization and excitation along the particle track even by uniform irradiation. This characteristic microscopically non-uniform dose delivery is expected to induce complex DNA damage and mutagenesis, in contrast to relatively uniform dose delivery in gamma-rays or electron beam irradiation. To investigate the distinct biological effects of heavy ions, especially to determine the effects of occupational and environmental exposure of very low doses of heavy charged particles, for example, since virtually no cells receive more than one traversal of cosmic ray HZE particle in its lifetime in a spaceflight environment, one approach is to select cells with the desired exposures from a randomly irradiated population. Using conventional track segment irradiation methods and sophisticated ion track detecting techniques, the position of the target cells and the ion tracks can be measured together. However, this conventional approach is not practical because all responses of many cells which do not contribute to the aim of the irradiation experiment must be measured. The alternative is to control each ion hit so that irradiation experiment is not a random Poisson process. A heavy ion microbeam can be used to selectively irradiate individual cells which can be analyzed afterward to determine what changes have occurred to that cell and to its un-irradiated neighboring cells. A heavy ion microbeam can also be used to look for pathways other than DNA damage, e.r., damage to the cell membrane or cytoplasm. We want to design and test a high-energy microbeam apparatus and a micron-resolution solid state detector for space radiobiology studies. In addition, we will develop in vitro models relevant to radiation risk using a microbeam capable of delivering individual charged particles to individual cells in situ. The system will allow us to critically determine the response of human cells to the single-particle traversals typically en-countered in space environmental exposures. During long-term space flight mission, its estimated that virtually no cell receives more than one Fe ion traversal in 3-year Mars mission scenario. Thus the use of the microbeam will aim to produce data for direct input into the analysis of human health risks during long-term space flights exposures involving exposure to low fluences of charged particles. A single-ion microbeam facility comprises a number of elements arranged to deliver reliably counted numbers of ions to a chosen biological target. The elements are: a) a source of ions of the appropriate energy, b) a means of limiting the location of the ions to an area less than the area of the target, c) a means of locating and moving the biological targets to the beam position, d) a means of detecting each ion as it traverses the target, and e) a means of shutting off the beam after the arrival of the chosen number of ions. A principal objective of this project is to develop and demonstrate a high resolution silicon detector, which will be able to determine the position of impact of energetic heavy ions in single cell radiation effects studies to within ~1 micron. An additional objective is a conceptual design of a heavy ion microbeam in the energy range up to 3 GeV. The beam will be collimated to ó~10-20 microns, to a region of one or very few cells. The microbeam will be implemented in a separate project at the NASA Space Radiation Laboratory (NSRL) in BNL, previously known as Booster Accelerator Facility(BAF). These developments will advance significantly the state-of-the art of high energy heavy ion microbeams and of high resolution heavy ion detectors. For the cell studies employing these tools, the necessary infrastructure will include a micropositioning stage with a microscope and auxilliary detectors. (2) Results: A new concept for a silicon detector with micron position resolution for heavy ions has been developed. Second iteration detectors have been completed and testing performed. A detector design with 30µm pitch has achieved a position resolution of 0.46µm rms for 1GeV/n Fe ions. The detectors also demonstrated very good energy resolution which enables simultaneous measurements of the impact position and ionization of a single heavy ion. Basic design studies for the heavy ion microbeam in the energy range of several hundred MeV (and up to 3 GeV) at the BNL NSRL have been completed.(Installation of the microbeam will have to be funded separately.) (3) Implications: The micron resolution detector, together with the microbeam, will be able to localize the position of an ion impact within a particular region of the cell. This is essential for studies in space radiobiology. (4) Synergisms: Radiation Effects Program, in particular, damage to central nervous system from radiation exposure. The objectives and specific aims of this project have been defined in consultation and with the agreement of the team working on radiation effects studies. (5) Milestones: The milestones for the project have been completed.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The use of microbeam provides a unique way to control the number of particles traversing individual cells and localizing the dose within the cell. High energy heavy charged particles transfer their energy to biological organisms through high density ionization and excitation along the particle track even by uniform irradiation. This characteristic microscopically non-uniform dose delivery is expected to induce complex DNA damage and mutagenesis, in contrast to relatively uniform dose delivery in gamma-rays or electron beam irradiation. To investigate the distinct biological effects of heavy ions, especially to determine the effects of occupational and environmental exposure of very low doses of heavy charged particles, for example, since virtually no cells receive more than one traversal of cosmic ray HZE particle in its lifetime in a spaceflight environment, one approach is to select cells with the desired exposures from a randomly irradiated population. Using conventional track segment irradiation methods and sophisticated ion track detecting techniques, the position of the target cells and the ion tracks can be measured together. However, this conventional approach is not practical because all responses of many cells which do not contribute to the aim of the irradiation experiment must be measured. The alternative is to control each ion hit so that irradiation experiment is not a random Poisson process. A heavy ion microbeam can be used to selectively irradiate individual cells which can be analyzed afterward to determine what changes have occurred to that cell and to its un-irradiated neighboring cells. Although the characteristic biological effects of heavy ions are supposed to be linked to the induction of high-LET-specific DNA lesions, other pathways of radiation effects are still interesting to consider, and there have been several reports that radiation effects of heavy ions may be transmitted from irradiated cells to neighboring unirradiated cells. A heavy ion microbeam can be used to look for pathways other than DNA damage, e.r., damage to the cell membrane or cytoplasm.

Task Progress:	The project's final year, the emphasis was on the evaluation of the micron resolution detectors. A second iteration of the silicon detectors was produced. Two beam tests were carried out in June and Septeber of 2004. The results of the beam tests and other lab tests are summaried below. 1. Charge Carrier Spread The extend of the charge carrier spread at the pixel plane is a key parameter in the optimization of the silicon detector. In order to achieve good position resolution and linear position response, the pitch of the pixel strips has to be smaller than that of the FWHM of the charge spread function. In previous measurements with red laser beams, we have observed that the charge spread is a very strong function of the total charge deposited by the ionization. The charge spread is also much greater than calculated value based solely on the diffusion process. The additional charge spread is attributed to an expansion process driven by the repulsion force of the high charge carrier density created by the ionization. The FWHM of the charge spread under a 1GeV/n Fe beam is about 40µm. This enables us to design detectors with pixel pitch of 30µm with good performance. 2. Position Resolution Signal to noise ratio measurements on the detector indicate that submicron position resolution can be achieved with this type of detector. However, since there is no other electronic detector system with better position resolution to define the beam, the measurement was done by stacking two identical detectors in the beam and measure the residual errors between the two devices. To reduce the parallax error from the beam's multiple scattering angles, the two detectors were placed 0.5mm away. The best position resolution for the 1GeV/n Fe ions is measured to be 0.46µm rms from a detector with 30µm pixel pitch. 3. Energy Resolution The stripixel detectors also have very good energy resolution for heavy ions. During a heavy ion run at NSRL, the beam was heavily fragmented. The detectors were able to measure the total ionization from each ion's impact. And the histogram of the de/dx measurement clearly resolves every ion species from iron down to nitrogen. The energy resolving capability of the detector allows the species as well as the impact position of an ion to be determined in real time.
Bibliography Type:	Description: (Last Updated: 10/10/2006)
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